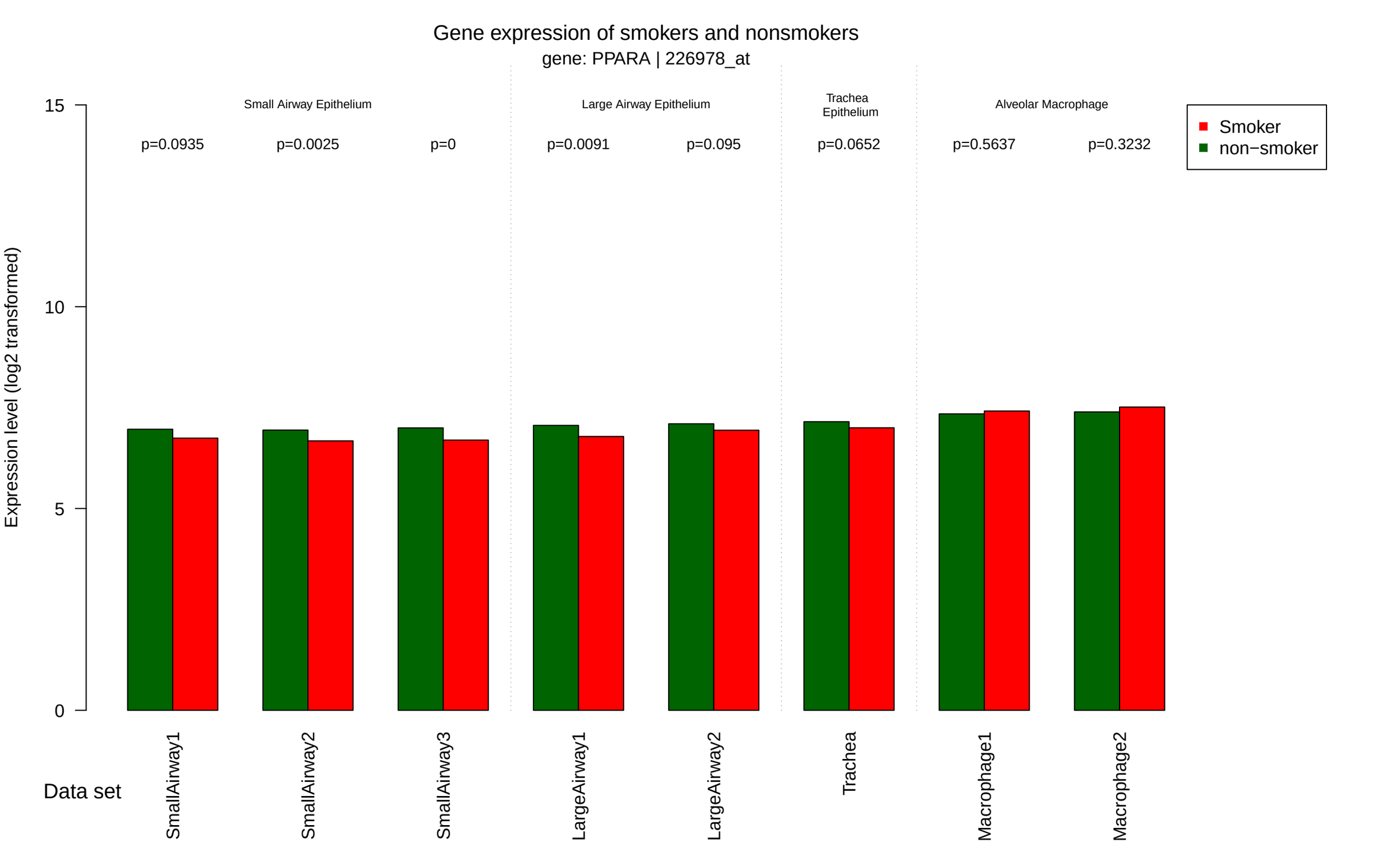
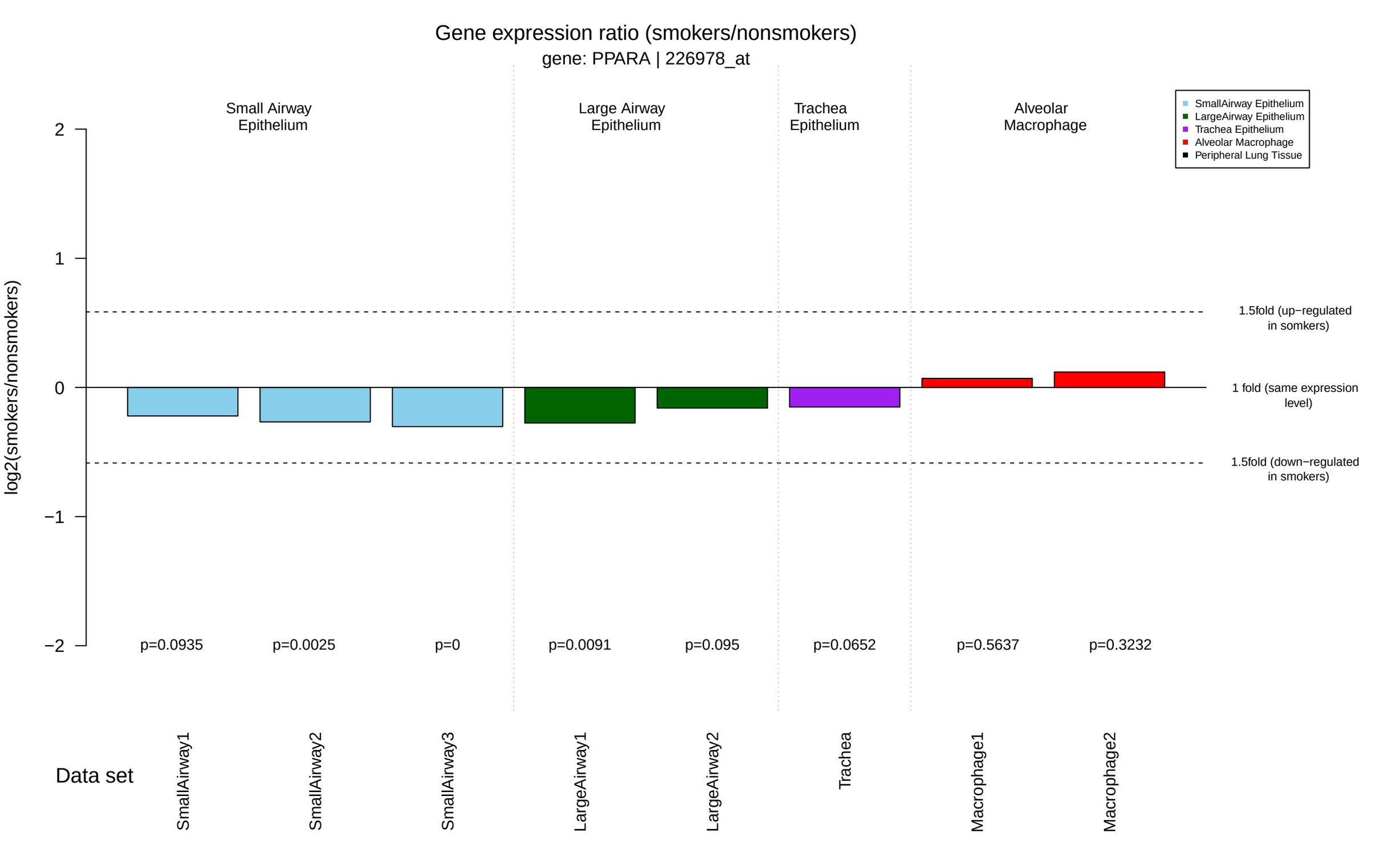
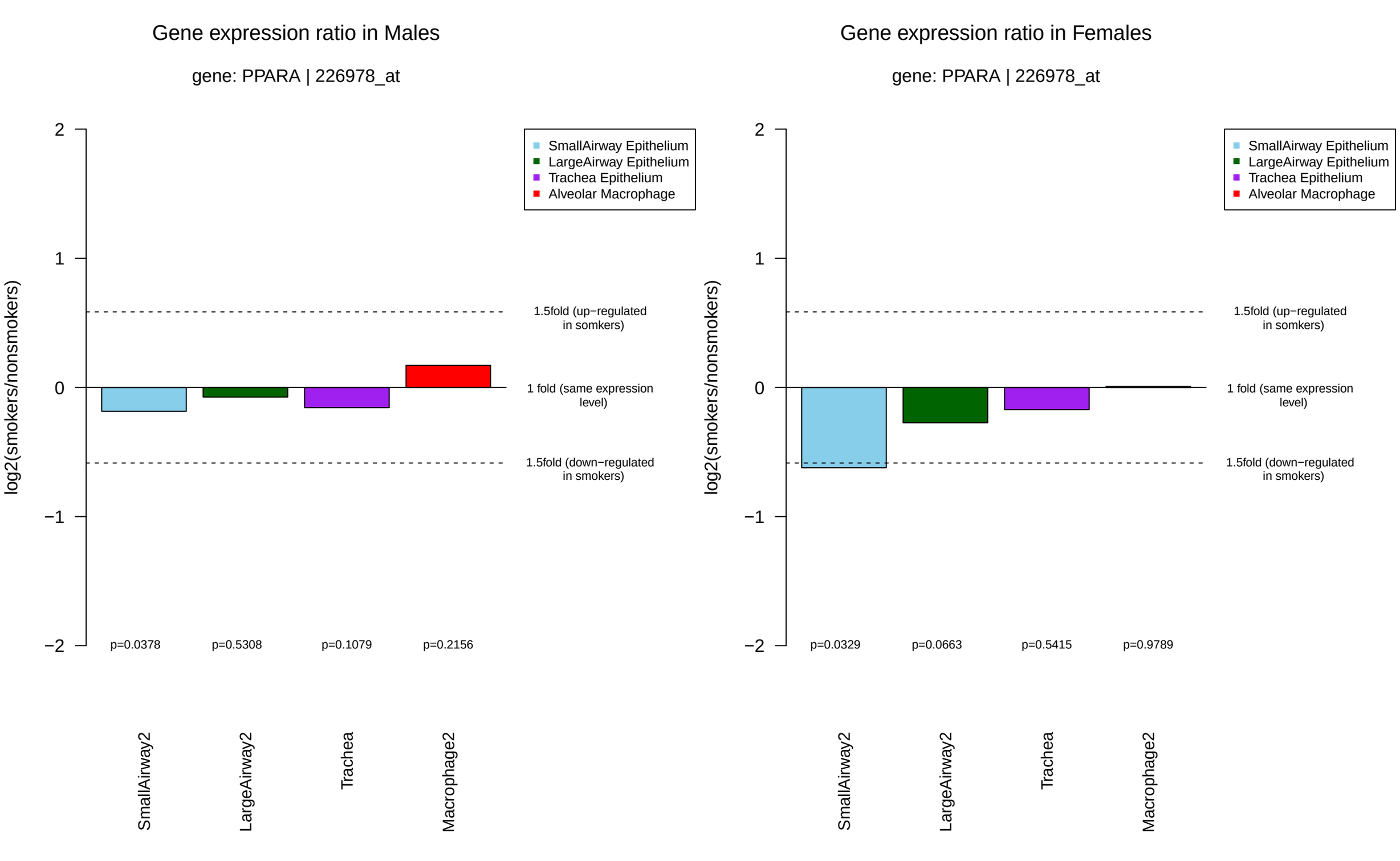
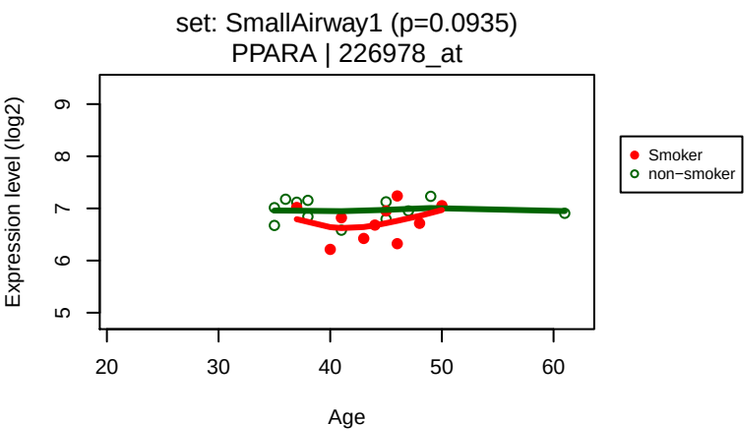
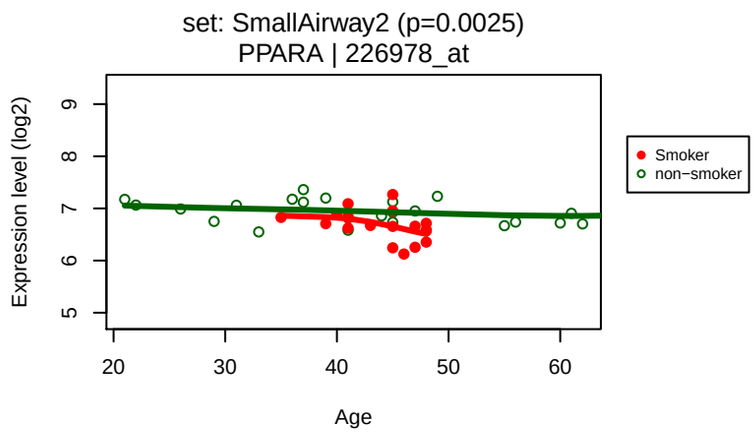
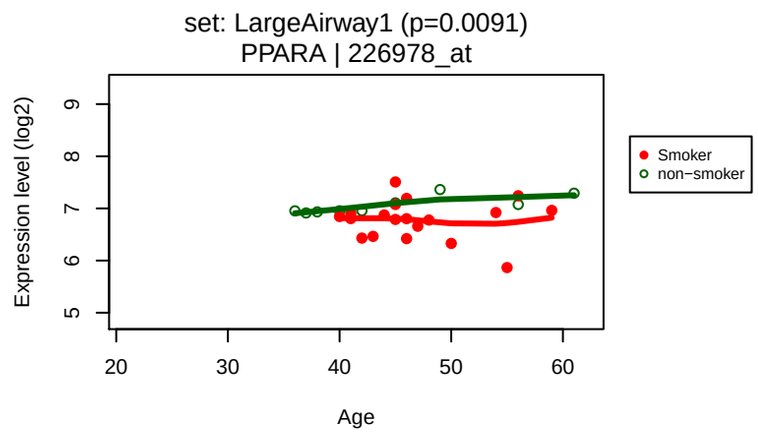

Gene expression of smokers and nonsmokers
gene: PPARA | 226978_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0935
p=0.0025
p=0
p=0.0091
p=0.095
p=0.0652
p=0.5637
p=0.3232
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

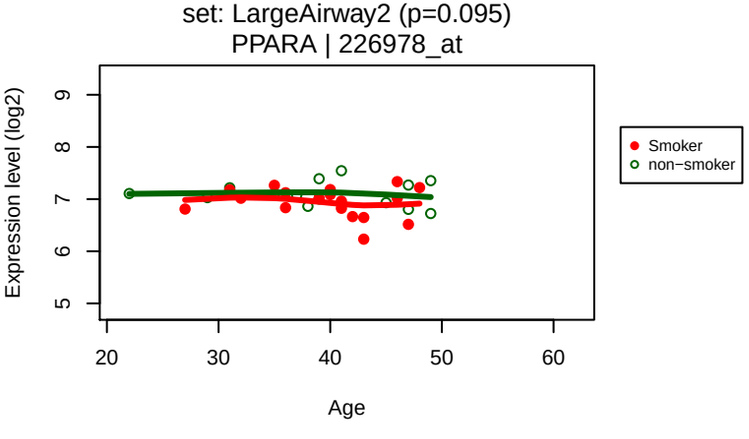

Gene expression ratio (smokers/nonsmokers)
gene: PPARA | 226978_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0935
p=0.0025
p=0
p=0.0091
p=0.095
p=0.0652
p=0.5637
p=0.3232
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

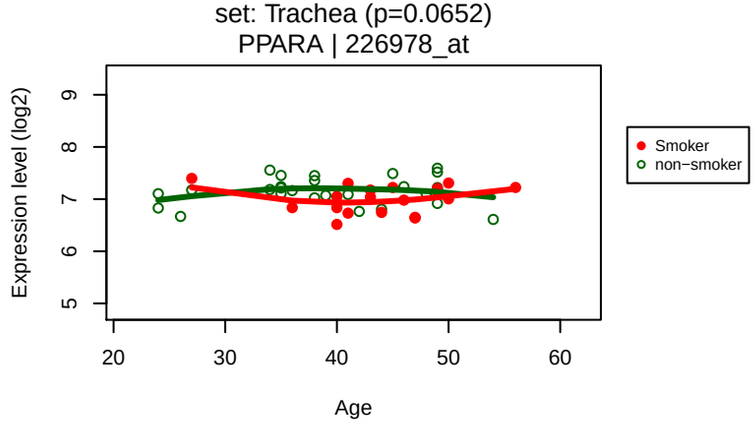

Gene expression ratio in Males
gene: PPARA | 226978_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0378
p=0.5308
p=0.1079
p=0.2156
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PPARA | 226978_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0329
p=0.0663
p=0.5415
p=0.9789
SmallAirway2
LargeAirway2
Trachea
Macrophage2

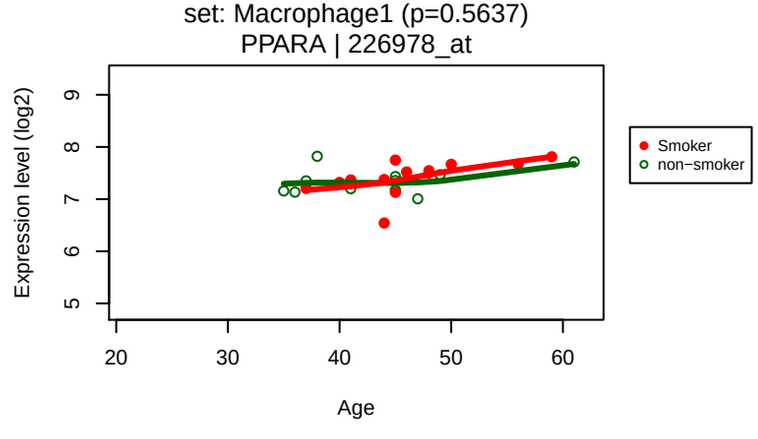

set: SmallAirway1 (p=0.0935)
PPARA | 226978_at
Expression level (log2)
Age
Smoker
non−smoker

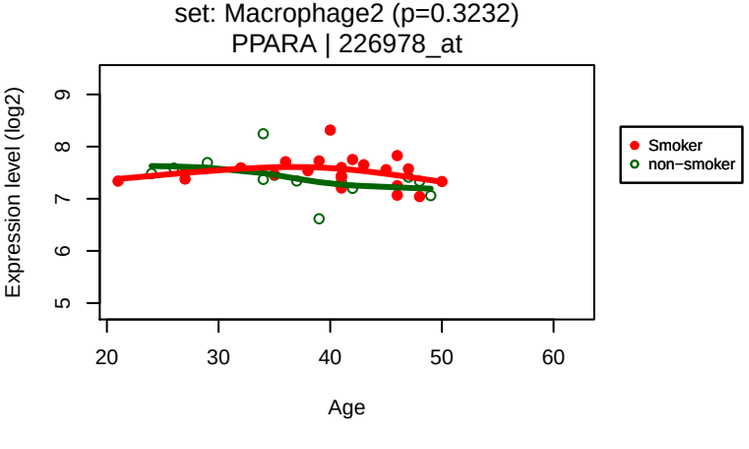

set: SmallAirway2 (p=0.0025)
PPARA | 226978_at
Expression level (log2)
Age
Smoker
non−smoker

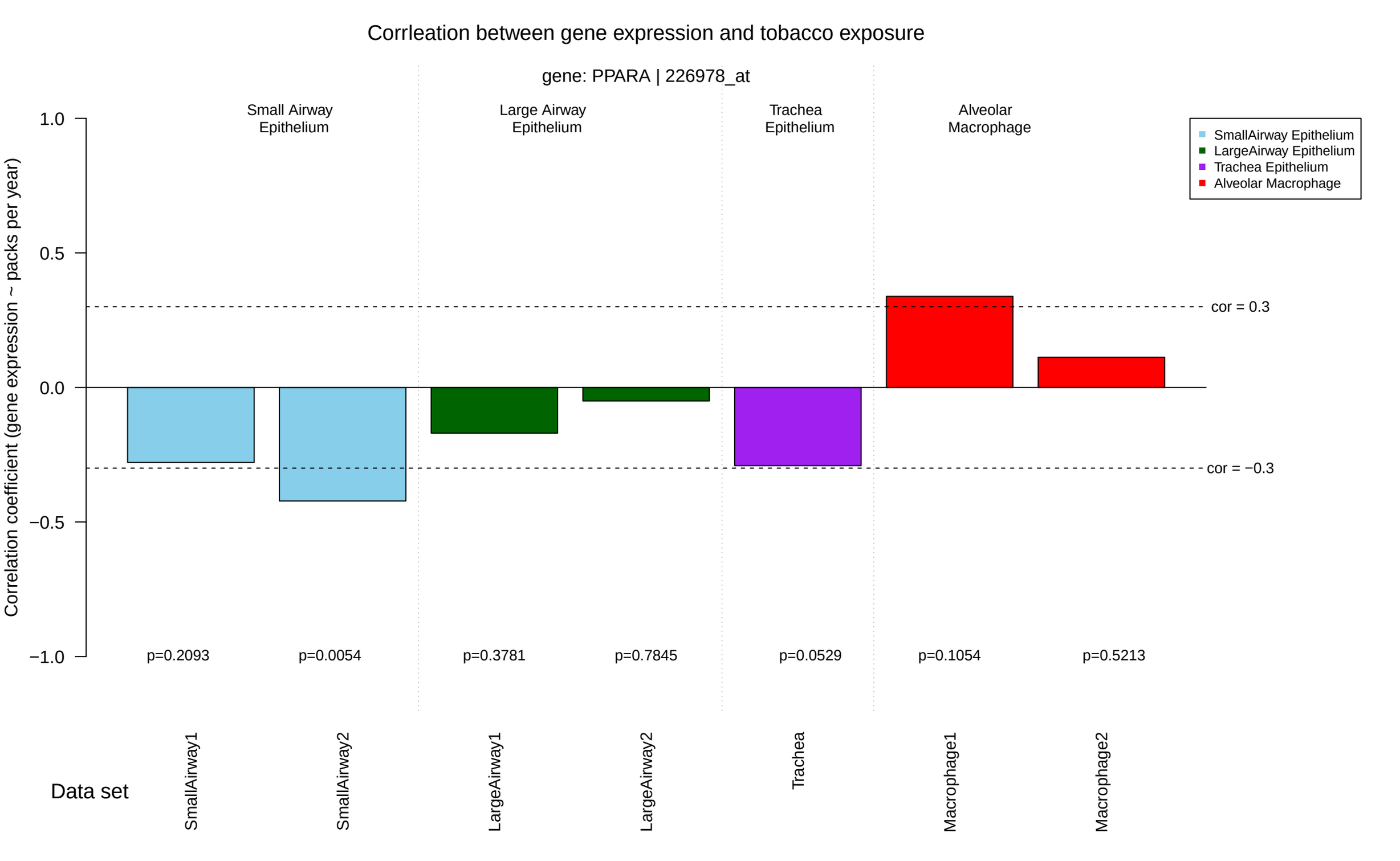

set: LargeAirway1 (p=0.0091)
PPARA | 226978_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.095)
PPARA | 226978_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0652)
PPARA | 226978_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5637)
PPARA | 226978_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3232)
PPARA | 226978_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PPARA | 226978_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2093
p=0.0054
p=0.3781
p=0.7845
p=0.0529
p=0.1054
p=0.5213
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2